Linux Package Management, Services, Logs & Processes

YUM & DNF | service & systemctl | Package Logs | ps, top & kill

Class notes - continuing after LVM. Commands explained one by one, with ROOT marked where a root shell is needed.

What This Set Covers

1. **Package Management** - installing, searching, updating and removing software with YUM (legacy) and DNF (modern)
2. **Managing Services** - starting, stopping and checking background services with `service` and `systemctl`
3. **Package Logs** - checking what the package manager did in `/var/log`
4. **Processes** - finding process IDs with `ps`, watching live activity with `top`, and stopping processes with `kill`

Commands are written without `sudo`. Any command tagged ROOT must be run from a root shell (or as the root user).

1. Package Management (YUM & DNF)

Simple idea: think of your phone's app store. You search for an app, tap install, and the store downloads it, sets it up, and later updates or removes it. A **package manager** is exactly that, but for Linux software. A **package** is one app or tool, packed into a single file (on Red Hat family systems, an `.rpm` file).

1.1 YUM vs DNF - the legacy command and its replacement

YUM (Yellowdog Updater, Modified) is the original package manager of the Red Hat family - RHEL, CentOS and Fedora. It is the **legacy** tool: standard on **RHEL / CentOS 7 and older**.

DNF (Dandified YUM) is the modern replacement. It became the default from **RHEL / CentOS 8 onwards** (and Fedora). It is faster, resolves dependencies better, and uses less memory.

The good news: **for the common operations covered here (search, install, update, remove), the commands are identical** - just swap `yum` for `dnf`. In fact, on RHEL 8/9 the `yum` command still exists - it is just a link that silently runs DNF.



1.2 Searching for a package

`yum search nginx`

Searches the repositories for packages whose name or description matches "nginx". Use this when you don't know the exact package name. No root needed - searching only reads information. (`dnf search nginx` on modern systems.)

1.3 Installing a package

`yum install nginx` ROOT

Downloads the nginx package and everything it depends on, then installs it. YUM/DNF shows the download size and asks for a `y` confirmation before proceeding. (`dnf install nginx` on modern systems.)

1.4 Updating packages

`yum update` ROOT

Updates **every** installed package to the newest version available in the repositories. Run it regularly to get security fixes.

`yum update nginx` ROOT

Updates only the named package (here, nginx) instead of the whole system. Note: it may also update other packages that the new version depends on.

1.5 Removing (uninstalling) a package

`yum remove nginx` ROOT

Uninstalls the package. It lists exactly what will be removed and asks for confirmation, so read the list before typing `y`. `dnf autoremove` ROOT additionally cleans up dependencies that are no longer needed by anything.

1.6 YUM vs DNF - same jobs, two names

Job	Legacy (RHEL 7)	Modern (RHEL 8/9)
Search for a package	`yum search nginx`	`dnf search nginx`
Install a package	`yum install nginx` ROOT	`dnf install nginx` ROOT
Update everything	`yum update` ROOT	`dnf update` ROOT
Remove a package	`yum remove nginx` ROOT	`dnf remove nginx` ROOT
Extra: list installed packages	`yum list installed`	`dnf list installed`
Extra: details of one package	`yum info nginx`	`dnf info nginx`

2. Managing Services (service & systemctl)

Simple idea: a **service** (also called a **daemon**) is a program that runs in the background with no window, quietly doing its job all the time - like kitchen staff in a restaurant. You never see them, but food keeps coming out. `sshd` is the SSH service: it sits in the background waiting for remote login connections.

2.1 The legacy way: the service command

`service` is the older command from the SysV init era (RHEL 6 and earlier generations). It still works on many current systems - on systemd-based distros it usually forwards your request to `systemctl`, but the exact behaviour depends on the distro and the service.

`service sshd start` ROOT

Starts the sshd service now - the machine begins accepting SSH connections.

`service sshd stop` ROOT

Stops the sshd service - existing sessions may drop and new SSH logins are refused.

`service sshd status`

Shows whether the service is running or stopped, plus its PID and recent log lines. Reading status usually needs no root.

`service sshd restart` ROOT

Stops then starts the service in one step. Used after editing a configuration file so the service reloads with the new settings.

2.2 The modern way: systemctl

Red Hat family systems from RHEL 7 onwards use **systemd** to manage services, and `systemctl` is its command (other systemd-based distros such as current Ubuntu work the same way). This is what you will actually use day to day - and what interviews ask about.

`systemctl start sshd` ROOT

Starts the service right now.

`systemctl stop sshd` ROOT

Stops the service right now.

`systemctl status sshd`

Shows running state, PID, memory use and the latest log lines - the first command to reach for when a service misbehaves.

`systemctl restart sshd` ROOT

Stop + start in one command, to apply new configuration.

`systemctl enable sshd` ROOT

Tells the system to **start sshd automatically at every boot**. It does NOT start the service right now - it only creates the auto-start link for next boot.

`systemctl disable sshd` ROOT

Removes the auto-start link so sshd will NOT start by itself at boot. It does NOT stop a service that is currently running, and the service can still be started manually or pulled in by another service that depends on it.

Lab safety: stopping or restarting `sshd` on a machine you are connected to over SSH can cut off your own session. Before practising, make sure you have an independent recovery path - local or VM console access, not another SSH session - and validate config edits with `sshd -t` ROOT before restarting.

Exam and interview favourite: `enable` and `start` are two different ideas. `enable` = "start automatically at boot" (a setting for the future). `start` = "run it now". To do both together in one step: `systemctl enable --now sshd` ROOT.

2.3 Running now vs starting at boot - two separate questions

active (running)
controlled by `start` / `stop` / `restart`
check with `systemctl status sshd`

enabled (auto-start at boot)
controlled by `enable` / `disable`
check with `systemctl is-enabled sshd`

A service can be **enabled but stopped** (will start at next boot, not running now), or **running but disabled** (running now, won't start by itself after reboot). Real servers usually need both: `start` it and `enable` it.

Distro naming: the SSH service is called `sshd` on Red Hat family systems (RHEL, CentOS, Fedora) and `ssh` on Debian/Ubuntu. Same service, different unit name.

3. Checking Package Logs

Simple idea: a **log** is the system's diary. Every time packages are installed, updated or removed, the package manager writes an entry - what happened, when, and whether it worked. When someone asks "who installed this, and when?", the log answers.

3.1 Where logs live

`cd /var/log`

Most classic system logs are stored under the `/var/log` directory - including the package logs below. Go here first, then `ls` to see the log files. One exception: systemd's own journal may be stored under `/run/log/journal` instead (lost on reboot) unless persistent journaling is enabled.

3.2 The DNF / YUM log files

`cat dnf.log`

The main DNF activity log (while inside `/var/log`): every install, update and remove transaction with timestamps. Reading log files in `/var/log` may need ROOT on some systems.

`cat dnf.rpm.log`

The lower-level RPM log: one line per individual package that was installed, upgraded or erased inside each transaction. Useful when you want package-by-package detail rather than the overall transaction.

`cat yum.log`

On legacy YUM systems (RHEL/CentOS 7), the equivalent history lives in **`/var/log/yum.log`** instead.

`tail -f dnf.log`

Watches the log live - new lines appear on screen as they are written. Handy while an install is running. Press **`Ctrl+C`** to stop watching.

Faster than reading raw logs: `dnf history` prints a tidy table of past transactions (installs, updates, removals) with IDs, and `dnf history info 5` shows the full detail of transaction number 5.

4. Processes: ps, top and kill

Simple idea: a **process** is any program currently running - every open app, every background service, even your shell. Each one gets a unique **PID** (Process ID), like an employee badge number. To manage a process you first find its badge number, then act on it.

4.1 ps - snapshot of processes

`ps`

Shows only processes owned by your user (same effective UID) and attached to your current terminal. Small and quick, but a very limited view.

`ps aux`

Shows **every process on the system**, for every user, with details: `a` = all users, `u` = user-oriented format (owner, %CPU, %MEM), `x` = include background daemons with no terminal. This is where you read the **PID** column (2nd column).

Spelling note: the classic form is `ps aux` with **no dash** (BSD-style options). It is often typed as `ps -aux` in class notes, but the correct, warning-free form is `ps aux`. To find one program quickly, filter it: `ps aux | grep sshd` (the grep process itself may show up in the results - a cleaner alternative is `pgrep -a sshd`, which prints just the matching PIDs and names).

4.2 top - the live task manager

`top`

Opens a live, self-refreshing view of processes sorted by CPU usage - the Linux equivalent of Task Manager's Processes tab. It keeps running until you quit.

Useful keys while top is running:

Key	What it does
`k`	**Send a signal to a process** - top asks for the PID, then the signal (press Enter for the default 15, SIGTERM). SIGTERM is a request: a process can catch or ignore it, and you can only signal your own processes unless you are root.
`d` or `s`	**Change the refresh rate** - top asks for the delay in seconds between screen updates (default is 3). Lower = faster updates.
`P`	Sort by CPU usage (the default).
`M`	Sort by memory usage - find what is eating RAM.
`q`	Quit top and return to the shell.

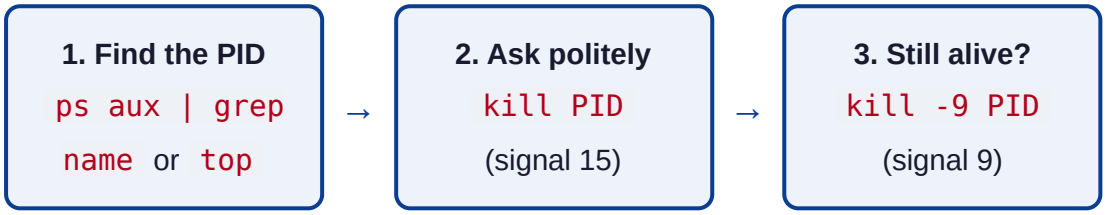
4.3 kill - stopping a process by PID

`kill 2451`

Sends the default signal **SIGTERM (15)** to PID 2451 - a polite "please shut down" that lets the process save data and exit cleanly. Killing another user's processes needs ROOT.

`kill -9 2451`

Sends **SIGKILL (9)** - the force kill. A process cannot catch or ignore it, so it ends without any chance to clean up. Use only when a normal `kill` didn't work, because unsaved data can be lost. Even SIGKILL cannot instantly remove a process stuck in uninterruptible sleep (state D) or an already-dead zombie process - those clear only when the stuck I/O finishes or the parent process reaps them.



5. Command Chart - Everything in One Place

Packages

Command	Use
`yum search nginx`	Search repositories for a package (legacy RHEL 7; `dnf search` today)
`yum install nginx` ROOT	Install a package and its dependencies
`yum update` ROOT	Update all installed packages (`yum update nginx` for just one)
`yum remove nginx` ROOT	Uninstall a package
`dnf list installed`	List every installed package
`dnf autoremove` ROOT	Remove leftover dependencies nothing needs anymore
`dnf history`	Show past install/update/remove transactions

Services

Command	Use

`service sshd start`
`service sshd stop`
`service sshd restart`
ROOT
	Legacy way to control a service - run one at a time (usually forwards to systemctl on systemd-based distros)

`service sshd status`	Legacy way to check if a service is running
`systemctl start sshd` ROOT	Start the service now
`systemctl stop sshd` ROOT	Stop the service now
`systemctl restart sshd` ROOT	Restart the service (apply new config)
`systemctl status sshd`	Check state, PID and recent logs
`systemctl enable sshd` ROOT	Auto-start at every boot (does NOT start it now)
`systemctl disable sshd` ROOT	Remove auto-start at boot (does not stop it, and it can still be started manually)
`systemctl enable --now sshd` ROOT	Enable and start in one step

Logs

Command	Use
`cd /var/log`	Go to the directory holding most classic system logs
`cat dnf.log`	Read DNF's transaction log (modern systems)
`cat dnf.rpm.log`	Read the per-package RPM detail log
`cat yum.log`	Read the package log on legacy YUM systems
`tail -f dnf.log`	Watch the log live as new lines are written

Processes

Command	Use
`ps`	Show your user's processes on this terminal
`ps aux`	Show every process with owner, CPU, memory and PID
`ps aux | grep sshd`	Find one program's PID quickly (grep itself may appear; `pgrep -a sshd` is a cleaner alternative)
`top`	Live task-manager view; `k` kill, `d` / `s` refresh rate, `q` quit
`kill 2451`	Politely stop PID 2451 (signal 15, SIGTERM)
`kill -9 2451`	Force-stop PID 2451 (signal 9, SIGKILL) - last resort

6. Interview Q&A

1. What is the difference between YUM and DNF?

Both are package managers for the Red Hat family, and their commands are almost identical. YUM is the legacy tool, standard up to RHEL/CentOS 7. DNF is the modern replacement from RHEL/CentOS 8 onwards - faster, with better dependency resolution. On RHEL 8/9, the yum command still exists but simply links to DNF.

2. How do you install, update and remove a package?

Install: `dnf install nginx`. Update everything: `dnf update`, or one package with `dnf update nginx`. Remove: `dnf remove nginx`. All three need root. On older systems, the same commands start with yum instead of dnf.

3. What is the difference between systemctl enable and systemctl start?

`start` runs the service right now but changes nothing about boot - whether it comes back after a reboot depends on whether the service is enabled. `enable` only creates the link that auto-starts the service at every boot - it does not start it now. For both effects at once, use `systemctl enable --now sshd`.

4. What is the difference between the service and systemctl commands?

`service` is the legacy SysV init command from older releases; on systemd-based distros it usually forwards the request to systemctl (exact behaviour varies by distro and service). `systemctl` is the native systemd command used on RHEL 7+ and other systemd-based distros such as current Ubuntu, and it adds boot control (enable/disable) that service never had.

5. How do you check if a service like sshd is running?

`systemctl status sshd` shows whether it is active or inactive, its PID, memory use and the latest log lines. To check only the boot setting, use `systemctl is-enabled sshd`.

6. Where does Linux keep logs, and how do you check package activity?

Classic system logs live under `/var/log` (systemd's journal may instead use the volatile `/run/log/journal`). Package activity on RHEL-family systems is in `dnf.log` (transactions) and `dnf.rpm.log` (per-package detail); legacy systems use `yum.log`. A tidier option is `dnf history`, which lists past install, update and remove transactions.

7. How do you find the PID of a running program?

Run `ps aux` to list every process, and filter it with grep: `ps aux | grep nginx` - the PID is the second column. Inside `top`, the PID is the first column of the live list.

8. What is the difference between kill and kill -9?

A plain `kill PID` sends SIGTERM (signal 15): a polite request that lets the process save data and exit cleanly. `kill -9 PID` sends SIGKILL, which a process cannot catch or ignore - it ends with no cleanup, so unsaved data can be lost. Even so, processes stuck in uninterruptible sleep (state D) or zombies are not removed instantly. Always try the plain kill first and use -9 only when the process refuses to die.

9. How do you kill a process without leaving top, and change how fast top refreshes?

Press `k` inside top, type the PID, and press Enter to accept the default signal 15. To change the refresh rate, press `d` (or `s`) and type the delay in seconds - the default is 3. Press `q` to quit top.

7. Sources

These notes follow the official documentation below. When in doubt, the man pages on your own system (`man systemctl`, `man ps`, ...) are the final word.

Topic	Source
DNF package management (RHEL 9)	https://docs.redhat.com/en/documentation/red_hat_enterprise_linux/9/html/managing_software_with_the_dnf_tool/index
Managing services with systemd	https://access.redhat.com/documentation/en-us/red_hat_enterprise_linux/7/html/system_administrators_guide/chap-managing_services_with_systemd
systemctl start/stop/enable/disable semantics	https://www.man7.org/linux/man-pages/man1/systemctl.1.html
service command (legacy)	https://manpages.debian.org/bookworm/init-system-helpers/service.8.en.html
Journal storage (/var/log vs /run/log/journal)	https://www.man7.org/linux/man-pages/man5/journald.conf.5.html
ps options and process selection	https://www.man7.org/linux/man-pages/man1/ps.1.html
top interactive keys	https://www.man7.org/linux/man-pages/man1/top.1.html
SIGTERM / SIGKILL behaviour	https://www.man7.org/linux/man-pages/man7/signal.7.html